

BALAVANTH

Technical Support Analyst | AI Builder

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PROFESSIONAL SUMMARY

Detail-oriented Technical Support Analyst at Tata Consultancy Services (TCS) with proven experience in geospatial ticket mapping, operational data management, and maintaining zero-shrinkage production targets. Consistently delivers 93% productivity and 100% utilization across high-accuracy data workflows. Independently builds production-grade AI systems — including RAG pipelines, autonomous agents, and full-stack applications — using Python, FastAPI, LangChain, and LLM APIs, demonstrating strong self-initiative and engineering capability beyond the core role.

WORK EXPERIENCE

Systems Analyst— IT Support & Mapping — Tata Consultancy Services (TCS) Aug 2024 – Present

- Execute high-volume geospatial ticket mapping using Apple's internal mapping tool (analogous to ArcGIS), processing location data with zero-shrinkage accuracy across production batches.
- Maintain consistent 93% productivity rate and 100% utilization across daily mapping and data verification workflows in a structured SLA-driven environment.
- Perform quality checks on mapped data to identify and resolve discrepancies, ensuring output meets Apple's internal accuracy and completeness standards.
- Manage ticket queues and track task progress through defined escalation channels to maintain uninterrupted pipeline throughput.
- Collaborate within a structured operations team to meet daily output targets while adhering to strict data handling and confidentiality protocols.

AI & TECHNICAL PROJECTS

VAULTIQ — Multi-Tenant RAG Knowledge Base | Python · FastAPI · FAISS · LangChain · HuggingFace · Vercel

- Engineered a production-deployed multi-tenant knowledge retrieval system using FAISS vector search and LangChain, enabling secure, context-aware Q&A over enterprise documents.
- Built FastAPI backend on HuggingFace Spaces with React frontend on Vercel — fully live and publicly accessible.

PRISIM — Multi-Source AI Research Agent | Python · LangGraph · ChromaDB · DuckDuckGo · Groq API

- Developed an autonomous research agent using LangGraph that retrieves, ranks, and synthesises information from multiple live sources with persistent vector memory.

KOBRA — Local Voice AI Assistant | Python · Porcupine · faster-whisper · Groq · edge-tts

- Built an offline-capable Windows voice assistant with custom wake-word detection, speech-to-text transcription, and LLM-driven spoken responses.

WAYWARD — Cross-Platform Mobile App | React Native · Expo · Supabase · Groq API

- Delivered a full-stack mobile application integrating real-time Supabase database and Groq LLM inference, debugging SDK and Python version compatibility issues end-to-end.

MARKETCREW — Multi-Agent Content Automation | CrewAI · SambaNova API · Python

- Designed a CrewAI multi-agent pipeline that autonomously researches, drafts, and publishes content using role-based agent orchestration.

TECHNICAL SKILLS

Support & Operations: Ticket Management, Geospatial Data Mapping, Quality Assurance, SLA Adherence, Data Verification, Escalation Management

AI & Automation: LangChain, LangGraph, CrewAI, FAISS, ChromaDB, Groq API, SambaNova API, RAG Systems, Agentic Workflows, Prompt Engineering

Programming: Python, JavaScript, TypeScript

Frameworks & Stack: FastAPI, Django, React, Supabase

DevOps & Deployment: Docker, Vercel, HuggingFace Spaces, Render, Git, GitHub

Operating Systems: Windows, Linux (Ubuntu)

EDUCATION

Bachelor of Computer Applications (BCA) — Bengaluru 2021 – 2024

CGPA: 7.8

12th Standard (PUC) — Seshadripuram PU College, Bengaluru 2019 – 2021

76%

10th Standard (SSLC) — St. Anne's High School, Bengaluru 2019

88%